Prof Ram Meghe Institute of Technology & Research , Badnera- Amravati.44701
(An Autonomous Institute)
Mid Semester Examination (MSE)/ Mid Semester Improvement Examination (MSIE)

Name of Examination :	Second Year B.Tech Winter-2024 Semester-III	
Name of Program :	Department of AI & DS .	
Name of Subject :	Discrete Mathematics	Time : 1 hr30 min
Subject Code :	3SA01	Maximum Marks :30

Instructions to Examinees :

1) Assume suitable data whenever necessary and clearly state the assumption made.
2) Use of mobile /blue tooth device/headphone/programmable calculators is strictly prohibited during Examination.
3) Write answers only using the blue/black pens.
4) Draw sketeches/diagrams/circuit diagrams neatly and cleanly.

		Marks	CO
Q.1.	a) Explain with example: i) Tautology ii) Contradiction iii) Contingency	(9)	1
	b) Show that the following Equivalences. i) $(\sim P \wedge (\sim Q \wedge R)) \vee (Q \wedge R) \vee (P \wedge R) \Leftrightarrow R$ ii) $P \vee (P \wedge Q) \Leftrightarrow P$	(6)	1
	OR		
Q.2.	a) What is compound statement ? Explain it's logical connectives.	(8)	1
	b) Find Conjunctive normal form $\sim \underbrace{(P \vee Q)}_{p} \leftrightarrow \underbrace{(Q \wedge P)}_{q}$.	(7)	1
Q.3.	a) Show that $A \rightarrow \sim D$ follows from the premises $A \rightarrow (C \vee B), B \rightarrow \sim A$, $D \rightarrow \sim C$.	(7)	2
	b) Symbolize the following statements:- i) Every apple is red. ii) Nobody likes cheat. iii) Some dogs are black.	(9)	2
	OR		
Q.4)	a) Explain the following :- i) Universal Quantifiers. ii) Existential Quantifiers.	(8)	2
	b) Show that $(\exists x)(P(x) \wedge Q(x)) \rightarrow (\exists x)P(x) \wedge (\exists x)Q(x)$	(7)	2

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Name of Examination: **Second Year B. Tech Winter-2024 Semester - III**
Name of Program: Artificial Intelligence and Data Science
Name of Subject: Data Structures — Time: **1 hr 30 min**
Subject Code: 3SA02 — Maximum Marks: **30 marks**

Instructions to Examinees:

1. Assume suitable data wherever necessary and clearly state the assumptions made.
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4. Draw sketches/ diagrams/ circuit diagrams neatly and cleanly.

		Marks	CO
Q.No 1 a	Explain the concept of data structure in detail.	05	1
Q.No 1 b	Define the terms: Insertion, Deletion, Traversing, Searching, Sorting	05	1
Q.No 1 c	Consider the following strings: **S= 'ABCDEF'** **T= 'GHIJKL'** Determine: 1] INSERT (T, 3, 'YZ') 2] INSERT (S, 3, 'YZ') 3] DELETE (S, 2, 2) 4] DELETE (T, 2, 2) 5] REPLACE (S, 'CD', 'MN')	05	1
	OR		
Q.No 2 a	Consider the T = 'aaba' and P = 'ab' Determine the INDEX of P using first pattern matching algorithm.	05	1
Q.No 2 b	Define the terms Substring and Indexing with example.	05	1
Q.No 2 c	What is an algorithm? Define with an example.	05	1
Q.No 3 a	Define a linear array and explain its memory representation.	05	2
Q.No 3 b	Define the steps involved in binary search algorithm.	05	2
Q.No 3 c	Discuss the pointer array with an example.	05	2
	OR		
Q.No 4 a	Define the steps involved in traversing a linear array.	05	2
Q.No 4 b	Discuss the steps involved in insertion operation in a linear array.	05	2
Q.No.4 c	What is a record structure? Provide an example.	05	2

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Name of Examination: **Second Year B. Tech Winter-2024 Semester - III**
Name of Program: **Artificial Intelligence and Data Science**
Name of Subject: **Programming Methodology** Time: 1 hr 30 min
Subject Code: **(3SA03)** Maximum Marks: **30 marks**

nstructions to Examinees:

1. Assume suitable data wherever necessary and clearly state the assumptions made.
2. Use of mobile phone/ blue tooth device/ headphone/ programmable calculator is strictly prohibited during the examination.
3. Write answers only using blue/ black pen.
4. Draw sketches/ diagrams/ circuit diagrams neatly and cleanly.

		Marks	CO
Q.No 1 a	What is JVM? What is byte code?	05	CO1
Q.No 1 b	Explain for loop with example.	05	CO1
Q.No 1 c	Explain the structure of Java program.	05	CO1
	OR		
Q.No 2 a	What is an operator? List out types of operator?	05	CO1
Q.No 2 b	Write a program to generate Fibonacci series: 1 1 2 3 5 8 13 21 34 55 89.	05	CO1
Q.No 2 c	Explain the difference between break and continue statement.	05	CO1
Q.No 3 a	Write a program to pass an integer number from command prompt and generate its factorial number.	05	CO2
Q.No 3 b	Explain types of array. How to declare Single dimensional array and how to initialize it?	05	CO2
Q.No 3 c	Describe the concept of constructor with example.	05	CO2
	OR		
Q.No 4 a	Write a class student having data members name, Roll No. and branch of a student. Also declare the two methods i.e. getdata( ) and display( ) for taking the input and display the same. Write a Java code for display the information for single student	05	CO2
Q.No 4 b	How variable are declare and initialized in Java?	05	CO2
Q.No.4 c	Explain the types of Constructors in Java? Write a Java program to find the Area of Circle using Constructor.	05	CO2

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Name of Examination: **Second Year B. Tech Winter-2024 Semester - III**
Name of Program: AI& DS
Name of Subject: Analog & Digital Devices
Subject Code: 3SA04

Time: 1 hr 30 min
Maximum Marks: **30 marks**

Instructions to Examinees:

1. Assume suitable data wherever necessary and clearly state the assumptions made.
2. Use of mobile phone/ blue tooth device/ headphone/ programmable calculator is strictly prohibited during the examination.
3. Write answers only using blue/ black pen.
4. Draw sketches/ diagrams/ circuit diagrams neatly and cleanly.
5.

		Marks	CO
Q.No 1a	Describe P-N Junction diode theory in detail	05	1
Q.No 1b	Interpret working of P-N junction diode in forward & reverse bias	05	1
Q.No 1c	Explain process of breakdown of diode due to i) Avalanche effect 2) zener effect.	05	1
	OR		
Q.No 2a	Explain the following terms i) knee voltage2) Reverse resistance 3) static resistance 4) Reverse saturation current.	05	1
Q.No 2b	Explain effect of temperature on V-I characteristics of P-N function diode.	05	1
Q.No 2c	Draw V-I chara. Of zener diode.List applications of zener diode.	05	1
Q.No 3a	What are 3 different configurations of transistor? What are advantages of CE configuration over CB?	05	2
Q.No 3b	Explain the significance of current gain α & β in transistor using various current components.	05	2
Q.No 3c	Explain the mechanism of current flow in N-P-N Transistor.	05	2
	OR		
Q.No 4a	The emitter current I_E in a transistor is 4mA,if the leakage current I_{CBO} is 6mA and α=.98 .Find collector and base current	05	2
Q.No 4b	Sketch a family of CE input characteristics for a transistor. Explain the shape of the curves qualitatively	05	2
Q.No.4c	Explain why $I_{CBO} << I_{CEO}$	05	2

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Name of Examination: **Second Year B. Tech Winter-2024 Semester - III**
Name of Program: **Artificial Intelligence and Data Science**
Name of Subject: **(Open Elective – 1 (C)) Introduction to Constitutional Law**
Time: **1 hr 30 min**
Subject Code: **(3SAOE1C)** Maximum Marks: **30 Marks**

Instructions to Examinees:

1. Assume suitable data wherever necessary and clearly state the assumptions made.
2. Use of mobile phone/ blue tooth device/ headphone/ programmable calculator is strictly prohibited during the examination.
3. Write answers only using blue/ black pen.
4. Draw sketches/ diagrams/ circuit diagrams neatly and cleanly.

		Marks	CO
Q.No 1 a	What do you mean by Constitution. Explain in brief General Idea about the Constitution	07	CO1
Q.No 1 b	Write Preamble of Constitution of India. Explain Four Main Ingredients of the Indian Preamble	08	CO1
	OR		
Q.No 2 a	Explain Salient Features of the Constitution of India.	07	CO1
Q.No 2 b	List of Important Articles in Indian Constitution and Explain any two parts of Articles in Indian Constitution.	08	CO1
Q.No 3 a	Explain relationship between Fundamental Rights and Human Rights.	07	CO2
Q.No 3 b	What do you mean by Fundamental Rights? Explain its meaning, nature & significance.	08	CO2
	OR		
Q.No 4 a	How many Fundamental Rights are there in the Indian Constitution? Explain in brief	07	CO2
Q.No 4 b	What do you mean by Law? Explain Judicial Review article in Indian Constitution.	08	CO2